

Exercise # 13.7

Question No 5

Find an equation for the tangent plane and parametric equations for the normal line to the surface at the point P.

$$x^2 - xyz = 56; \quad P(-4, 5, 2)$$

$$\nabla F = (2x - yz, -xz, -xy), \text{ so } n = (-18, 8, 20),$$

So the tangent plane is given by $-18x + 8y + 20z = 152$, normal line $x = -4 - 18t$,

$$y = 5 + 8t, \quad z = 2 + 20t$$

Question No 6

$$z = x^2 + y^2; \quad P(2, -3, 13)$$

At P, $\partial z / \partial x = 4$ and $\partial z / \partial y = -6$,

tangent plane $4x - 6y - z = 13$, normal

$$\text{line } x = 2 + 4t, \quad y = -3 - 6t, \quad z = 13 - t$$

Question No 7

$$Z = 4x^3y^2 + 2y; \quad P(1, -2, 12)$$

At P , $\partial z / \partial x = 48$ and $\partial z / \partial y = -14$,
tangent plane $48x - 14y - z = 64$,

normal line $x = 1 + 48t$, $y = -2 - 14t$

$$z = 12 - t$$

Question No 8

$$Z = \frac{1}{2}x^7y^{-2}; \quad P(2, 4, 4)$$

At P , $\partial z / \partial x = 14$ and $\partial z / \partial y = -2$

tangent plane $14x - 2y - z = 16$

normal line $x = 2 + 14t$, $y = 4 - 2t$,

$$z = 4 - t$$

Question No 9

$$Z = xe^{-y}; \quad P(1, 0, 1)$$

At P , $\partial z / \partial x = 1$ and $\partial z / \partial y = -1$

tangent plane $x - y - z = 0$,
normal line $x = 1 + t, y = -t, z = 1 - t$

Question No 10

$$Z = \ln \sqrt{x^2 + y^2}; P(-1, 0, 0)$$

At P , $\partial z / \partial x = -1$ and $\partial z / \partial y = 0$,
tangent plane $x + z = -1$, normal
line $x = -1 - t, y = 0, z = t$

Question No 29

Find parametric equations for the tangent line to the curve of intersection of the paraboloid $z = x^2 + y^2$ and the ellipsoid $x^2 + 4y^2 + z^2 = 9$ at the point $(1, -1, 2)$.

$$n_1 = 2i - 2j - k, \quad n_2 = 2i - 8j + 4k$$

$$n_1 \times n_2 = -16i - 10j - 12k \text{ is tangent}$$

to the line, $x(t) = 1 + 8t$, $y(t) = -1 + 5t$, $z(t) = 2 + 6t$

Question No 30:

Find the parametric equations for the tangent line to the curve of intersection of the cone $z = \sqrt{x^2 + y^2}$ and the plane $x + 2y + 2z = 20$ at the point $(4, 3, 5)$

$$f(x, y, z) = \sqrt{x^2 + y^2} - z, \quad n_1 = \nabla f(4, 3, 5) =$$

$$\frac{4}{5}i + \frac{3}{5}j - k, \quad n_2 = i + 2j + 2k,$$

$$n_1 \times n_2 = (16i - 13j + 5k)/5 \text{ is tangent}$$

$$\text{to the line, } x(t) = 4 + 16t, \quad y(t) = 3 - 3t$$

$$z(t) = 5 + 5t. \text{ The point } (4, 3, 5) \text{ lies on}$$

both surfaces.